The Mathematics of Dubner's Ball

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Abstract

A positive integer w is a *twin prime witness* if $6w-1$ and $6w+1$ are both prime. The set of witnesses $\mathcal{W}$ (OEIS A002822) inherits a rich additive structure: w is *decomposable* if $w = u + v$ for some $u, v \in \mathcal{W}$. The *Dubner Ball* is the directed graph whose nodes are the elements of $\mathcal{W}$ and whose edges connect each witness to its additive progenitor pairs.

We study the geometry of this graph: its residue-class decomposition into three active planes modulo 5, the natural polar coordinate system within each plane, and the orthographic 3D projection that reveals the full additive architecture of $\mathcal{W}$. We characterise five qualitatively distinct ball structures corresponding to the five logically consistent combinations of the Middle Number Conjecture (MNC), the Twin Prime Conjecture (TPC), and a bridging condition on the additive structure (BC). A visualisation programme for distinguishing these worlds computationally is described and implemented.

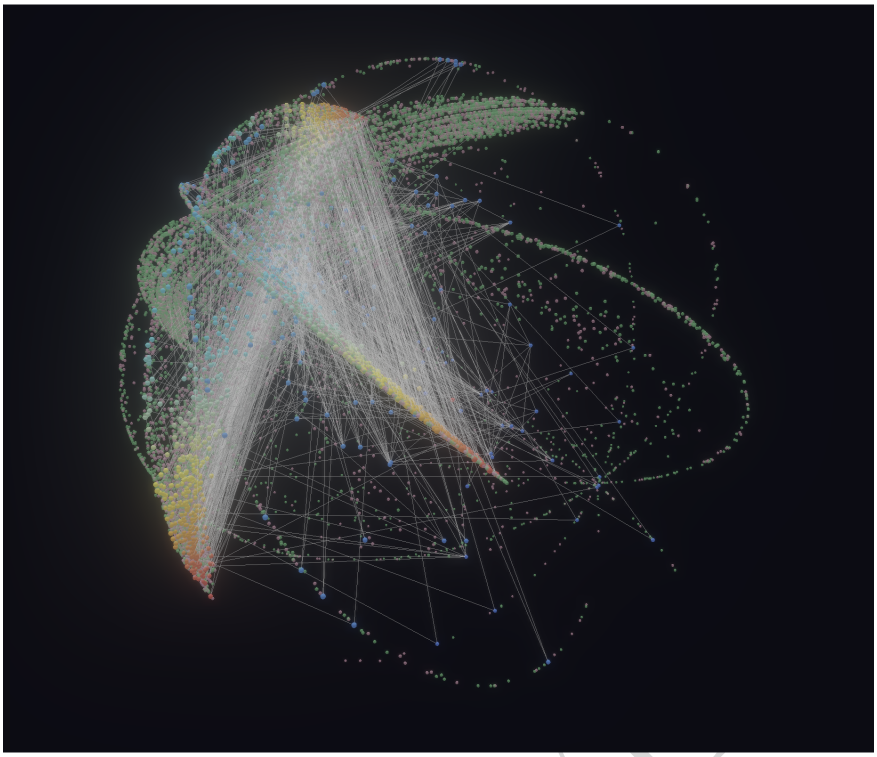

Figure 1: The Dubner Ball: twin prime witnesses rendered as a 3D point cloud, coloured by residue class modulo 5 ($k = 0$: yellow/green, $k = 2$: blue/cyan, $k = 3$: red/orange), with progenitor filaments connecting each witness to its progenitor pair. The ball is shown at $|\mathcal{W}| = 20{,}000$ witnesses. Witnesses with $k \in \{1, 4\}$ are absent by Lemma 3.1.

1 Introduction

A positive integer w is a *twin prime witness* if $6w-1$ and $6w+1$ are both prime. The set of witnesses $\mathcal{W}$ is the OEIS sequence A002822; its first few elements are $1, 2, 3, 5, 7, 10, 12, \ldots$ The *Twin Prime Conjecture* (TPC) asserts that $\mathcal{W}$ is infinite.

The witnesses carry an additive structure: some $w \in \mathcal{W}$ can be expressed as $w = u + v$ with $u, v \in \mathcal{W}$. The *exception set* $\mathcal{X} = \mathbb{N}_{\geq 1} \setminus (\mathcal{W} + \mathcal{W})$ collects the positive integers that admit no such decomposition. Computationally, $\mathcal{X}$ is known to contain the twelve-element set

$$\text{A243956} = \{1, 16, 67, 86, 131, 151, 186, 191, 211, 226, 541, 701\},$$

and no further elements have been found up to 2×10^{10} [3].

Harvey Dubner [3] investigated this structure in depth (following a suggestion of Stephen Wagler communicated via Paulo Ribenboim) and proposed two conjectures. The *Middle Number Conjecture* (MNC) asserts that every $w \in \mathcal{W}$ with $w > 1$ decomposes as a sum of two witnesses, i.e. $\mathcal{W} \cap \mathcal{X} = \{1\}$. The stronger *Middle Even Number Conjecture* (MENC) asserts that $\mathcal{X}$ is finite — equivalently, that $\mathcal{X}$ is a finite superset of A243956. Dubner's most precise conjecture, which we call Dubner 3, asserts that $\mathcal{X} = \text{A243956}$ exactly. The three conjectures are strictly ordered: Dubner 3 $\Rightarrow$ MENC $\Rightarrow$ MNC.

Definition 1.1 (Dubner Ball)**.** The *Dubner Ball* is the directed graph $G = (\mathcal{W}, E)$ where

$$E = \{(w, u, v) : u + v = w,\ u \leq v,\ u, v \in \mathcal{W}\}.$$

Each edge records a *progenitor pair* (u, v) for the node w.

The name honours Harvey Dubner (1928–2019), the electrical engineer and amateur mathematician whose investigations of the additive structure of twin prime witnesses directly inspired this visualisation. Section 8 records this debt in more detail.

Figure 1 (above) shows the Dubner Ball rendered as a 3D point cloud: each point is a witness, coloured by residue plane, with filaments connecting each witness to its progenitor pair. The three active planes ($k = 0, 2, 3 \pmod 5$) spiral outward as w grows, and the filament structure reveals the additive architecture at a glance.

The paper is organised as follows. Section 2 states the three conjectures MNC, BC, TPC and the five logically consistent worlds they define. Section 3 establishes the mod-5 residue constraint and the resulting three-plane decomposition. Section 4 introduces the polar coordinate system within each plane. Section 5 characterises the ball geometry in each world. Section 6 discusses the exception set $\mathcal{X}$. Section 7 outlines the progenitor graph statistics. Section 8 describes the visualisation programme.

2 Three Conjectures and Five Worlds

We work with three conjectures about the witness set $\mathcal{W}$.

Conjecture 2.1 (Twin Prime Conjecture, TPC)**.** The set $\mathcal{W}$ of twin prime witnesses is infinite.

Conjecture 2.2 (Middle Number Conjecture, MNC)**.** Every witness $w \in \mathcal{W}$ with $w > 1$ is the sum of two witnesses: there exist $u, v \in \mathcal{W}$ with $u + v = w$.

MNC is due to Dubner [3], following a suggestion of Stephen Wagler passed to Dubner by Ribenboim [4]. In Dubner's original formulation, a *middle number* is an integer sandwiched between a twin prime pair, i.e. $6w$ for $w \in \mathcal{W}$; his conjecture is that every middle number greater than 6 is the sum of two middle numbers, which is exactly MNC in our witness notation.

Conjecture 2.3 (Bridging Conjecture, BC)**.** For every threshold $V \geq 1$ there exist $u, v, w \in \mathcal{W}$ with $u \leq v \leq V < w$ and $u + v = w$.

BC asserts that no matter how far along $\mathcal{W}$ we have accumulated, some pair of accumulated witnesses sums to a *new* witness beyond the current frontier. It implies TPC: for any V, BC supplies a witness $w > V$, so $\mathcal{W}$ is unbounded.

The eight combinations of truth values for MNC, BC, TPC reduce to exactly five logically consistent *worlds* [1]:

World	MNC	BC	TPC	Status
1	T	T	T	conjectured actual world
4	T	F	F	$\mathcal{W}$ finite, MNC holds throughout
5	F	T	T	$\mathcal{W}$ infinite, orphan witnesses exist
7	F	F	T	$\mathcal{W}$ infinite, additive structure stalls at V^*
8	F	F	F	$\mathcal{W}$ finite with orphan witnesses

Worlds 2, 3, 6 are logically impossible: BC $\Rightarrow$ TPC (so BC true forces TPC true), and MNC $\wedge$ TPC $\Rightarrow$ BC (so MNC true and TPC true together force BC true).

3 Residue Classes and the Five-Plane Decomposition

For any $w \in \mathcal{W}$, the integers $6w-1$ and $6w+1$ are both prime. Since $6w \equiv 0 \pmod 6$, the residue class of w modulo 5 is constrained:

Lemma 3.1. *Every twin prime witness $w > 1$ satisfies $w \equiv 0, 2,$ or $3 \pmod 5$.*

Proof. If $w \equiv 1 \pmod 5$ then $6w+1 \equiv 7 \equiv 2 \pmod 5$ — not divisible by 5 — but $6w-1 \equiv 5 \equiv 0 \pmod 5$, so $5 \mid 6w-1$, forcing $6w-1 = 5$ (i.e. $w = 1$) or $6w-1$ composite. Similarly, if $w \equiv 4 \pmod 5$ then $6w+1 \equiv 25 \equiv 0 \pmod 5$, so $5 \mid 6w+1$, forcing $6w+1 = 5$ (impossible for $w \geq 1$) or $6w+1$ composite. Hence for $w > 1$ neither $w \equiv 1$ nor $w \equiv 4$ is compatible with both $6w \pm 1$ being prime. □

Definition 3.2 (Plane)**.** For $k \in \{0, 2, 3\}$, the *k-plane* is the subgraph of the Dubner Ball induced by witnesses $w \equiv k \pmod 5$.

The witness $w = 1$ lies in the $k = 1$ class and is the unique seed. All further witnesses lie in the $k \in \{0, 2, 3\}$ planes.

Remark 3.3. Progenitor edges can cross planes: if $u \equiv 2$ and $v \equiv 3 \pmod 5$, then $w = u + v \equiv 0 \pmod 5$. The plane decomposition partitions nodes but not edges.

4 Polar Coordinates Within Each Plane

Each plane inherits a natural polar coordinate system from the progenitor structure.

Definition 4.1 (Ball coordinates)**.** For a witness w with canonical progenitor pair (u, v), $u \leq v$ (choosing $u = \min\{u' : u' + v' = w,\ u', v' \in \mathcal{W}\}$), define:

$$\begin{aligned} r(w) &= \log\bigl(w - 1 + u/v\bigr), \\ \alpha(w) &= 2\pi \cdot u/v, \\ \psi(k) &= 2\pi k/5 \quad \text{(plane azimuth, } k = w \bmod 5). \end{aligned}$$

The three-dimensional Cartesian position of w in the ball is:

$$\mathbf{p}(w) = r\cos\alpha \cdot \mathbf{e}_1(\psi) + r\sin\alpha \cdot \mathbf{e}_z,$$

where $\mathbf{e}_1(\psi) = (-\sin\psi, \cos\psi, 0)$ is the in-plane radial direction and $\mathbf{e}_z = (0, 0, 1)$.

Remark 4.2. The angle $\alpha \to 0$ corresponds to highly lopsided progenitor pairs ($u \ll v$); $\alpha \to \pi$ to balanced pairs ($u \approx v$). Since the canonical choice always selects $u \leq v$, we have $u/v \leq 1$ and hence $\alpha \in (0, \pi]$, placing all points in the upper half-ball. The radial coordinate r grows logarithmically with w, so the ball expands without bound if and only if $\mathcal{W}$ is infinite (TPC).

5 The Five Worlds

Each of the five consistent worlds of Section 2 predicts a qualitatively distinct ball structure.

World 1: MNC true, BC true, TPC true

The conjectured actual world. $\mathcal{W}$ is infinite; every V has a bridging pair; every witness $w > 1$ decomposes as a sum of two smaller witnesses.

Ball character. The ball grows without bound across all three residue planes. Every node $w > 1$ has at least one incoming progenitor edge. The animation runs indefinitely, building outward with no dead zones. MENC predicts $\mathcal{X}$ is finite; Dubner 3 predicts $\mathcal{X} =$ A243956 exactly.

World 4: MNC true, BC false, TPC false

$\mathcal{W}$ is finite with largest element w^*.

Ball character. A finite, frozen sculpture. Every node $w > 1$ has an incoming progenitor edge (MNC holds), so the graph is internally well-connected below w^*. The exception set $\mathcal{X}$ is co-finite.

World 5: MNC false, BC true, TPC true

$\mathcal{W}$ is infinite and every V has a bridging pair (BC), but some witness w has no decomposition $w = u + v$ with $u, v \in \mathcal{W}$.

Ball character. The ball grows forever, but contains isolated nodes with no incoming edges — witnesses that appear in $\mathcal{W}$ but cannot be generated as sums of smaller witnesses. These *orphan nodes* are unreachable by any chain of additive decompositions from the seed. BC holds via a relay mechanism in which the decomposable witnesses collectively bridge every V, compensating for the gaps left by the orphans.

World 7: MNC false, BC false, TPC true

$\mathcal{W}$ is infinite (TPC holds), but some threshold V^* has no bridging pair: no $u, v \in \mathcal{W}$ with $u, v \le V^*$ sum to a witness above V^*. The additive closure of $\mathcal{W} \cap [1, V^*]$ under pairwise addition therefore cannot reach any witness beyond V^*.

Ball character. The ball built from progenitor edges below V^* is a finite frozen structure. Beyond V^* there exist witnesses in $\mathcal{W}$, but they have no progenitor path back through the ball: they are disconnected from the additive architecture below V^*. A companion visualisation renders these *ghost witnesses* — nodes of $\mathcal{W}$ that float above the frozen ball with no incoming edges from below.

World 8: MNC false, BC false, TPC false

$\mathcal{W}$ is finite. The exception set $\mathcal{X}$ is co-finite.

Ball character. A finite, frozen sculpture as in World 4, but with some nodes lacking incoming progenitor edges (MNC failures). The two finite worlds are visually indistinguishable unless MNC-failure nodes are explicitly marked.

6 The Exception Set

Definition 6.1 (Exception set)**.** The *exception set* $\mathcal{X}$ is the complement of the sumset $\mathcal{W} + \mathcal{W}$ in $\mathbb{N}_{\ge 1}$:

$$\mathcal{X} = \mathbb{N}_{\ge 1} \setminus (\mathcal{W} + \mathcal{W}).$$

It is known unconditionally that $\mathcal{X} \supseteq$ A243956, and computation confirms no further elements exist up to 2×10^{10}.

Remark 6.2. The three Dubner conjectures of Section 1 correspond to three assertions about $\mathcal{X}$:

- **MNC**: $\mathcal{X} \cap \mathcal{W} = \{1\}$ — no witness except 1 is non-decomposable.
- **MENC**: $\mathcal{X}$ is finite — equivalently, a finite superset of A243956.
- **Dubner 3**: $\mathcal{X} =$ A243956 exactly.

These are strictly ordered: Dubner 3 $\Rightarrow$ MENC $\Rightarrow$ MNC. The definition of $\mathcal{X}$ is robust: a new exception beyond 701 would falsify Dubner 3 and MENC but would merely enlarge $\mathcal{X}$, leaving MNC open.

Remark 6.3. In the ball picture, $\mathcal{X}$ is exactly the set of nodes with no incoming progenitor edge: $w \in \mathcal{X}$ if and only if there are no $u, v \in \mathcal{W}$ with $u + v = w$. In World 1, MENC predicts $\mathcal{X}$ is finite; Dubner 3 predicts $\mathcal{X} =$ A243956. In World 5, $\mathcal{X}$ is infinite (infinitely many orphan nodes). In Worlds 4 and 8, $\mathcal{X}$ is co-finite ($\mathcal{W}$ is finite so $\mathcal{W} + \mathcal{W}$ is finite).

7 Progenitor Graph Structure

The following statistics characterise the additive architecture of the ball and are studied computationally in `notebook/residue-progenitor.ipynb`.

1. **In-degree**: $\deg^-(w) = |\{(u, v) : u + v = w,\ u \leq v,\ u, v \in \mathcal{W}\}|$ — the number of distinct progenitor pairs for w.
2. **Depth**: $d(w) =$ length of shortest progenitor path from w back to the seed $\{1\}$.
3. **Branching entropy**: Shannon entropy of the in-degree distribution.
4. **Bottleneck witnesses**: those w with $\deg^-(w) = 1$, i.e. witnesses with a unique progenitor pair.

8 Visualisation Programme

The full visualisation programme is implemented in `notebook/dubner-ball/dubner-ball.ipynb`. It produces:

1. **Per-plane 2D polar plots** (Figure 2): one panel per residue class $k \in \{0, 2, 3\}$, plus a combined overlay. Colour encodes progenitor balance $\delta_{uv} = (v - u)/w$ (blue = balanced, red = lopsided). Filaments connect each node to its progenitors.
2. **3D orthographic projection** (Figure 3): all three active planes folded into 3D and projected onto a 2D screen at azimuth $\psi_{\text{view}} = \pi$ $(= \frac{1}{2} \cdot 2\pi)$, elevation $\phi_{\text{view}} = \pi/6$ $(= \frac{1}{12} \cdot 2\pi)$.
3. **Five Worlds panels**: synthetic renderings of each world produced by truncating or injecting nodes (figures to be added in a future revision).

[Figure: per-plane 2D polar plots — to be generated from notebook]

Figure 2: Per-plane 2D polar plots of the Dubner Ball at $|\mathcal{W}| = 10{,}000$. Panels left to right: $k = 0$ ($\psi = 0$), $k = 2$ ($\psi = 4\pi/5$), $k = 3$ ($\psi = 6\pi/5$), combined overlay.

[Figure: 3D orthographic projection — to be generated from notebook]

Figure 3: Orthographic projection of the Dubner Ball ($|\mathcal{W}| = 10{,}000$), viewed from azimuth π and elevation $\pi/6$.

Acknowledgements

Harvey Dubner. This paper would not exist without the work of Harvey Dubner (1928–2019). Dubner was an electrical engineer from New Jersey whose lifelong passion for prime numbers produced dozens of records and discoveries using purpose-built hardware of his own design. His 1984 "Dubner cruncher", built with his son Robert, was at the time competitive with supercomputers for large-number arithmetic, and by 1993 he was responsible for more than half of all known primes over 2,000 digits.

The conjecture at the heart of this paper — that every twin prime witness $w > 1$ is the sum of two twin prime witnesses — originates with Stephen Wagler, who suggested it to Paulo Ribenboim, who passed it to Dubner in October 1996. Dubner investigated it thoroughly, found the twelve exceptional witnesses now catalogued as OEIS A243956 (including the seed $w = 1$), and published his findings in the *Journal of Recreational Mathematics* [3]. His paper is a model of careful computational exploration in the service of genuine mathematical insight.

The author first encountered Dubner's conjecture in a Reddit post by user `u/Heavy-Sympathy5330` on `r/math` in May 2026, with the title "This conjecture is so underrated." The post reproduced a scan of Dubner's introduction and noted, with some justice, that the conjecture deserves a wider audience. That post set off the chain of thought that produced the present research programme; it is a pleasure to acknowledge it here. The Dubner Ball is named in Dubner's honour: he provided both the conjecture and the computational evidence that give the ball its structure.

The visualisation infrastructure uses the Three.js interactive viewer at `3d/dubner-ball/` and an accompanying Python notebook.

AI systems (Claude, Aristotle, ChatGPT, Gemini) contributed to the development of this paper. The nature and extent of that collaboration is documented in full in [2].

References

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[4] P. Ribenboim, *The New Book of Prime Number Records*, Springer, 1996.